

AWS CLI Overview

The AWS Management Console offers a user-friendly graphical interface for creating and managing resources. However, for automation and managing multiple resources efficiently, the AWS Command Line Interface (CLI) is a better choice. The CLI allows you to interact with AWS services programmatically through APIs, offering a more scalable solution for resource management.

Why Use AWS CLI?

1. **Automation:** AWS CLI is ideal for automating tasks and managing resources via scripts.
2. **Efficiency:** Command-line operations are faster for creating, managing, and terminating resources.
3. **Programmatic Access:** With CLI, you can invoke AWS APIs programmatically for advanced use cases.

Getting Started with AWS CLI

1. **Installation:**
 - AWS CLI is built with Python. To install it, search for “AWS CLI download” on the web or visit the official AWS website.
 - Follow the installation instructions provided.
2. **Configuration:**
 - After installation, configure AWS CLI using your credentials.
 - Open a terminal or command prompt and run `aws configure`.
 - Enter your Access Key ID, Secret Access Key, default region, and output format (e.g., `json`).

Example command:

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```
aws configure
```

Verify the configuration:

sql

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```
aws sts get-caller-identity
```

- 3.
4. **Creating Resources:**
 - To create and manage AWS resources, use CLI commands.
5. Examples:
 - **S3 Bucket:**
 - List buckets: `aws s3 ls`
 - Create a bucket: `aws s3 mb s3://bucket-name`

- Remove a bucket: `aws s3 rb s3://bucket-name`

EC2 Instance:

css

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```
aws ec2 run-instances --image-id ami-xxxxxxx --count 1
--instance-type t2.micro --key-name MyKeyPair --security-group-ids
sg-903004f8 --subnet-id subnet-6e7f829e
```

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6. Best Practices:

- **Avoid Using Root Credentials:** Always use IAM user credentials to avoid security risks.
- **Refer to Official Documentation:** For detailed command usage and options, refer to the official AWS CLI documentation.

Alternative Tools

- **CloudFormation and Terraform:** For more complex infrastructure setups, AWS CloudFormation and Terraform offer infrastructure as code solutions, allowing you to define your infrastructure in scripts and deploy it programmatically.

Summary

The AWS CLI is a powerful tool for managing AWS resources quickly and efficiently. It complements the AWS Management Console by providing automation and scriptable operations. Use the CLI for routine tasks and consider CloudFormation or Terraform for more complex setups.